Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #40 | Retail, E-Commerce & Finance
Student Startup Crowdfunding Platform

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1. Problem Context & Overview
A crowdfunding engine for university student ventures, supporting milestone-based funding targets,
backer reward tier selection, and verified fund releases upon faculty committee review.
Target Stakeholders / Actors: Student Founder, Campaign Backer

- **Functional Rrequirenment(FR)**

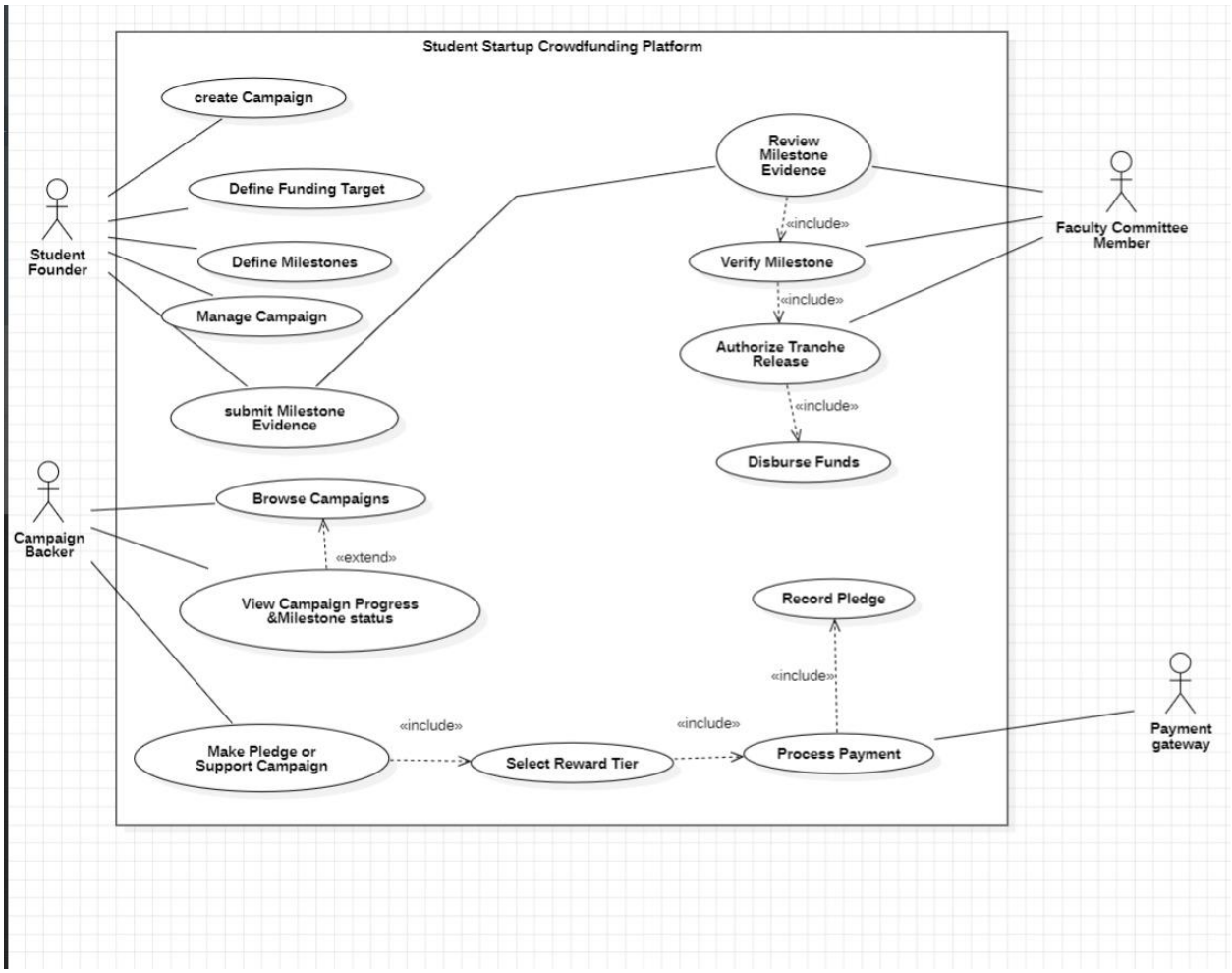
Req-ID	Type	Description	Priority	Accepatance criteria	Rationale (short)	Comments
FR-001	Functional	The system shall collect backer pledges and disburse funds to the student venture in tranches only upon verification of predetermined project milestones.	High	**Pass:** Milestone evidence is and the approved tranche is released. **Fail:** Funds are released without milestone verification.	Ensures funds are released only after successful milestone verification.	**Given requirement — do not change.**
FR-002	Functional	The system shall allow the Student Founder to create a campaign by defining the funding target and project milestones.	High	**Pass:** A campaign is created with a funding target and at least one milestone. **Fail:** Campaign cannot be created without the required details.	Enables student founders to establish and manage their crowdfunding campaigns.	Matches **Create Campaign, Define Funding Target, Define Milestones**
FR-003	Functional	The system shall allow Campaign Backers to browse campaigns, view campaign progress and milestone status, and make pledges by	High	**Pass:** Backer can view campaigns, select a campaign, view its progress, and select a reward tier before	Helps backers make informed funding decisions.	Matches **Browse Campaigns** and **View Campaign Progress & Milestone**

		selecting a reward tier.		pledging. **Fail:** Any required campaign or reward information is unavailable.		
FR-004	Functional	evidence.The system shall allow the Student Founder to submit milestone evidence and the Faculty Committee Member to review and verify the evidence	High	**Pass:** Evidence can be submitted and the Faculty Committee Member can review and verify it. **Fail:** Evidence cannot be submitted or verified.	Provides faculty oversight before funds are released.	Matches **Make Pledge → Select Reward Tier → Process Payment → Record Pledge**.
FR-OO5	Functional	The system shall process backer payments and record successful pledges after payment confirmation.	High	**Pass:** Successful payment results in a recorded pledge and confirmation. **Fail:** A failed payment is recorded as a successful pledge.	Ensures accurate and reliable pledge processing	Matches **Review → Verify → Authorize Tranche Release**

- **Non Functioanl Requirement (NFR)**

Req-Id	TYPE	Descrption	Priority	Accepatnces criteria	Rationale (short)	Comments
NFA-001	Non functional	The payment processing gateway must comply with PCI-DSS guidelines with tokenized payment authorizations.	High	**Pass:** Payment processing satisfies PCI-DSS requirements and uses tokenization. **Fail:** Required security standards are not satisfied	rotects sensitive payment informatio	**Given NFR — do not change.**
NFA-002	Non Functional	The system shall process key user requests within 3 seconds under normal operating load.	High	**Pass:** At least 95% of requests complete within 3 seconds. **Fail:** More than 5% take longer than 3 seconds.	Provides a responsive user experience.	New NFR based on performance requirement; measurable for testing.

- **Use case daigram**



Actors

1. **Student Founder**
2. **Campaign Backer**
3. **Faculty Committee Member**
4. **Payment Gateway**

Student Founder

- Create Campaign
- Define Funding Target
- Define Milestones
- Manage Campaign
- Submit Milestone Evidence

Campaign Backer

- Browse Campaigns
- View Campaign Progress & Milestone Status
- Make Pledge or Support Campaign
- Select Reward Tier

- Process Payment
- Record Pledge

Faculty Committee Member

- Review Milestone Evidence
- Verify Milestone
- Authorize Tranche Release
- Disburse Funds

Payment Gateway

- Process Payment

- **Use-Case Flow**

Use Case: Make Pledge or Support Campaign

Primary Actor: Campaign Backer

1. Preconditions

- The campaign is available on the platform.
- The campaign is accepting pledges.
- A reward tier is available.

2. Postconditions

- The pledge is recorded successfully.
- The selected reward tier is recorded.
- Payment confirmation is provided to the Campaign Backer.

3. Main Success Scenario

1. Campaign Backer browses available campaigns.
2. System displays the available campaigns.
3. Campaign Backer selects a campaign.
4. System displays campaign details and progress.
5. Campaign Backer selects **Make Pledge or Support Campaign**.
6. System displays available reward tiers.
7. Campaign Backer selects a reward tier.
8. Campaign Backer enters the pledge amount.
9. System sends the payment request to the Payment Gateway.
10. Payment Gateway processes the payment successfully.
11. System records the pledge.
12. System displays payment and pledge confirmation.

4. Alternate Flow — Payment Failure

13. Payment Gateway declines the payment.
14. System displays **Payment Failed**.
15. Campaign Backer can retry the payment.
16. If the Backer cancels, the pledge is not recorded